Habib University

Course Title: Digital Logic and Design (EE-172 / CS-130 / CE-222)

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Assignment No.: 03

Release Date: March 17th, 2025 **Due by:** March 26th, 2025, 11:59 PM

Total points: 100 **Points obtained:**

Student Name:	**Student ID:**	**Section:**

Purpose:

The purpose of the assignment is to help you analyze, design and minimize combinational logical circuits. In addition, this assignment will also aid you to practice the tools and techniques used to minimize logic functions.

Instructions:

1. This assignment can be done individually or in pairs (group of 2).
2. All questions should be answered in **black ink only**. (Extra sheets can be used)
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor/TA.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to answer correctly to the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work or improper submissions will result in a reduction of your points.

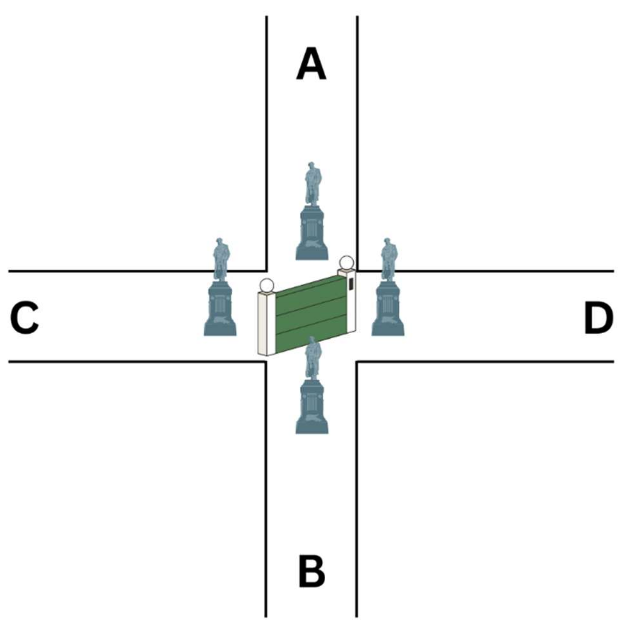
Late submission penalty:

1) 1-day late submission – 20% deduction of the maximum allowable marks
2) 2-days late submission – 40% deduction of the maximum allowable marks
3) No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply r-base (binary, octal, decimal, hexadecimal numbers systems) to digital systems and carry out arithmetic operations and conversions	
CLO 2	Apply principles of Boolean Algebra to represent and build equivalent realizations of digital logic (circuits)	
CLO 3	Design combinational logic circuits using logic gates	✓
CLO 4	Design sequential systems using finite state machine methodology	

P#	Questions	Pts
1	Design a combinational circuit that converts a 4-bit excess-3 number to BCD.	6
2	In a fantasy adventure game, a **treasure room** is protected by a magical gate that opens based on specific conditions. The room has four **guardian statues** (A, B, C, and D), each with a pressure plate in front of them. Players can step on these plates to activate them (logic HIGH), and the gate control system uses their states to decide when the treasure room opens. **Gate Logic:** • The treasure room gate opens if both statues C and D are activated. • The gate also opens if either C or D is activated, but A and B are not both activated at the same time. • If A and B are both activated, but C and D are not both activated, the treasure room remains locked. • The treasure room will also open if either A or B is activated while C and D are not both activated. • If no statues are activated, the treasure room remains locked. Using the sensor outputs A, B, C, and D as inputs, design a logic circuit to control the gate (G). The output of the circuit is G which is HIGH when the gate is open and LOW when the gate is closed. Implement and simplify the circuit as much as possible and show all steps.  Figure 01	12

3 You are required to design a digital circuit for detecting errors in transmission and reception using error detection code of parity check. **3x4+4**

a) Design odd parity generator module for transmitting 03-bit data input. Parity generator circuit receives three data bits (i.e., D_0 D_1 D_2) and generates Parity bit P as shown in Transmitter side of Figure 02. Provide truth table, Boolean expression and logic diagram for generating P.

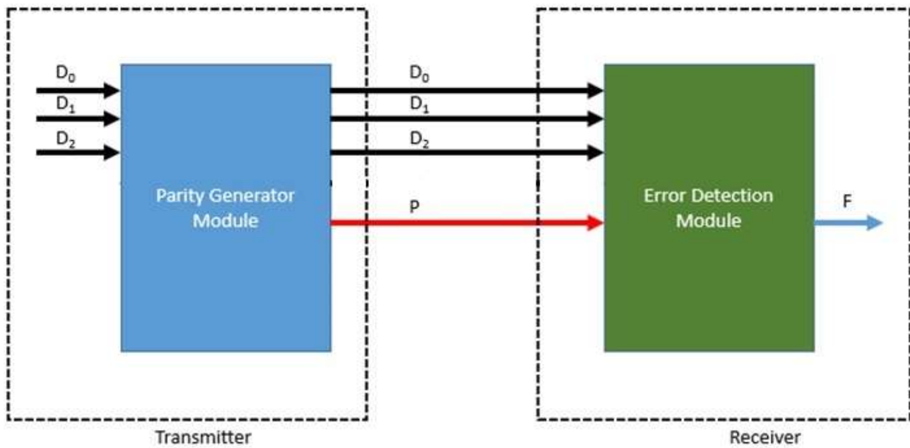


Figure 02

b) Design odd parity tester circuit which receives 04 bits (i.e., D_0 D_1 D_2 P) and generates a Flag bit F such that F = 1 when there is error in transmission and F = 0 when there was no error in transmission.
Provide truth table, Boolean expression, and logic diagram for generating F.

c) Provide XOR logic implementation of part (a) and (b)
Provide logic diagram for generating F.

d) Simulate NAND logic implementation of part (a) and (b) on any software.
Attach snapshot of three test cases
Test Case I: Parity bit generation by parity generation module
Test Case II: Flag generation by error detector module when there is no error
Test Case III: Flag generation by error detector module when there is error

e) Write conclusion on advantages and drawbacks of your transmission and reception system and suggest some ways to improve your system

4 Given the following circuit: **4x4**

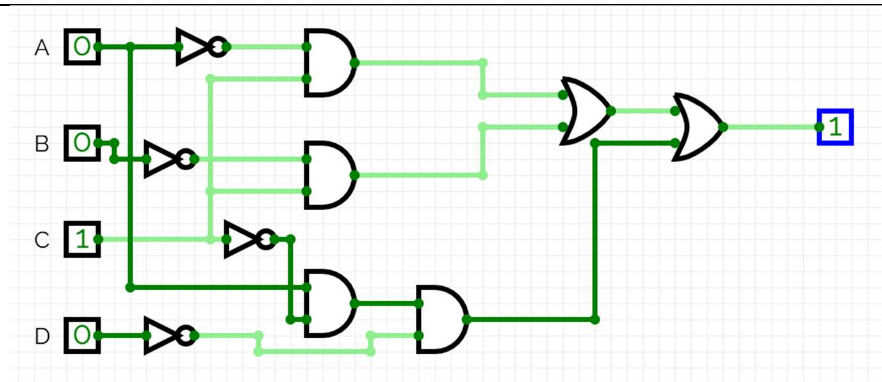


Figure 03

a) Run Analysis procedure i.e., find the Boolean representation of the logic circuit and obtain its truth table.
b) Using two 3x8 decoders, implement the expression you found in part a.
c) Using one 2x1 multiplexer and any combination of gates, implement the same expression you found in part a.
d) Using only 2x1 multiplexers and inverters, implement the same expression you found in part a.

5 Implement the following devices and show their circuit diagrams: 9x2

a) 2-bit by 4-bit multiplier using logic gates only. [To avoid cluttered work, make required adder/s first (show a proper schematic for that as well) and then use it as a block]
b) A full subtractor using 1x4 and 1x2 demultiplexers.

("For reference, read about binary subtractors from here: https://www.electronics-tutorials.ws/combination/binary-subtractor.html")

6 You have recently joined a **security company** that designs digital lock systems for vaults. The current system, which uses a **4-bit binary code to unlock the vault**, has been reported to have inconsistencies where some inputs result in multiple conflicting outputs. Your task is to **debug and correct the design**. 6x2

You ask the previous engineer for the **truth table** they have been using, and they provide you with the following data:

I1	I2	I3	I4	I5	I6	I7	I8	Unlock	Alarm
1	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	1
0	0	1	0	0	0	0	0	0	1
0	0	0	1	0	0	0	0	1	0
0	0	0	0	1	0	0	0	1	1
0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	0	1	0	1	0
0	0	0	0	0	0	0	1	1	1

a) You proceed to write the equations for all 4 outputs in terms of input and make the logic circuit (you can draw the circuit on paper or on any of the software).

<table>
<tr><td></td><td>b) Once you have the equations and the logic circuit you proceed to find the fault in the design and propose your own design and implementation of it. (Hence make the new truth table, and equations and for your design).</td><td></td></tr>
<tr><td>7</td><td>Mid-week is going on and the health of the students is deteriorating because of the stress. You decide to make a robot that roams around the campus and when it encounters a student who is currently having mids, it runs a check on their mental and physical health and informs the health center if the student is unwell. However, for this, the robot would first need to identify whether the student in front of them is currently having a mid or not. You will have to design a combinational circuit which detects from the student's id whether they are currently in the group who is having their mid or not.
• This year they are taking the mid in three phases. They take the student's id's modulus by 3 and divide the students according to the following scheme:
o Group 1: Remainder = 0
o Group 2: Remainder = 1
o Group 3: Remainder = 2
• The mid are carried out one by one for each of the above groups.
• Since right now the group that is having their mid is group 1, your robot would have to check up on the students whose ids modulus 3 is 0.
• Since Group 2 is next it will not be a problem if your robot checks up on a Group 2 student, too. Hence, you decide to keep the group 2 students as don't care conditions in your circuit.
a) Design a combinational circuit which caters to the above scenario for the ids from 0 to 15. Please show all the relevant working (the truth table, Standard form equations, K-map, Minimized algebraic equations, and the final circuit diagrams).
b) Now you realize that there can also be the case that the robot encounters two students at one time instead of 1. In that case you would need to prioritize which student to check first. You decide to prioritize the student with the greater id. Since you have already built the system for checking a student's id you would like to make a separate component that compares the ids and later integrate both components. Design a combinational circuit that compares two students' ids. Show all relevant steps
c) Now, you would need to join your circuits from (a) and (b) to complete your detection system. Instead of drawing the whole circuit, you can draw boxes labelled with (a) and (b). Just make sure that your circuit can deal with both the cases of 1 and 2 students appropriately and that both the components (a) and (b) are connected correctly.
d) You are planning to add more features to this robot for the final phase. Right now, you are just brainstorming different modifications. You decide to make your robot capable of dealing with more than 2 students at a time. What</td><td>2x6 + 2x4</td></tr>
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	modifications can you do to make your robot capable of comparing more than 2 IDs?	